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**Research Article**

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# Rangewide connectivity and migration ecology in *Limnothlypis swainsonii* (Swainson's Warbler) uncovered with barometric geolocators

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## ABSTRACT

Conserving migratory birds requires a detailed understanding of their full annual cycle, including nonbreeding distributions, migratory pathways, migratory phenology, and patterns of migratory connectivity. This information could be particularly important for managing *Limnothlypis swainsonii* (Swainson's Warbler), a Nearctic–Neotropical migrant of conservation concern. We tracked *L. swainsonii* from across their breeding distribution in the southeastern U.S. to identify migration routes and nonbreeding locations. In addition to traditional light-level geolocators, we used barometric pressure data collected by multisensor geolocators (pressure tags), which we combined with a global weather reanalysis and a movement model (GeoPressureR) to estimate precise, continuous locations, and collect previously unavailable flight behavior data. We deployed 43 pressure tags and 43 light-level geolocators on breeding male *L. swainsonii* across 6 states in 2021. We recaptured 31 (36%) tagged individuals in 2022, with 26 tags used for analysis. Light-level geolocators ($n = 12$) experienced chronic shading, resulting in broad nonbreeding estimates and little information during migration. Pressure tags ($n = 14$), however, provided accurate nonbreeding estimates (~10–40 km) and all stopover regions. We found a longitudinal divide in nonbreeding distributions: western breeders of the Mississippi Valley and Missouri Ozarks migrated across the Gulf of Mexico to spend the nonbreeding season in the Yucatan Peninsula; eastern breeders from the Appalachians and Atlantic Coastal Plains migrated to Cuba and the Bahamas, with en route stops in Florida and Georgia. Pressure tags also revealed precise details of migratory movements, including phenology, flight durations, and altitudes. Most notably, individuals traveled significantly faster and spent less time at stopovers during fall migration than in spring, antithetical to current understanding. Migratory pathways and wintering grounds of *L. swainsonii* suggest eastern and western breeding populations experience different environmental conditions, ecological and anthropogenic barriers, and habitat quality during migration.

**Keywords:** barometric geolocators, connectivity, *Limnothlypis swainsonii*, migration ecology, phenology

**How to Cite**

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## LAY SUMMARY

- It is critical to track migratory birds throughout their annual to better understand migratory connectivity and ecology.
- There is little information about the migration ecology of *Limnothlypis swainsonii* (Swainson's Warbler), a species of conservation concern.
- We deployed 86 barometric and light-level geolocators on *L. swainsonii* across their breeding range in 2021, recovered 31 geolocators (36%) in 2022, and used 26 tags for analysis.
- Whereas light-level data provided broad (100+ km) nonbreeding estimates, barometric data revealed more accurate estimates, as well as continuous stopover sites, exact migratory phenology, and novel flight data.
- We found *L. swainsonii* exhibits a longitudinal pattern of migratory connectivity: individuals from the west crossed the Gulf of Mexico to spend the nonbreeding season in the Yucatan Peninsula, while from the east, individuals migrated down peninsular Florida to the Caribbean.
- Data showed birds migrated faster in the fall than in the spring.

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## Conectividad a escala de todo el rango y ecología de la migración en *Limnothlypis swainsonii* reveladas mediante geo-localizadores barométricos


### RESUMEN

La conservación de aves migratorias requiere una comprensión detallada de su ciclo anual completo, incluyendo las distribuciones no reproductivas, las rutas migratorias, la fenología migratoria y los patrones de conectividad migratoria. Esta información podría ser particularmente importante para el manejo de *Limnothlypis swainsonii*, un migrante neártico–neotropical de interés para la conservación. Rastreamos a *L. swainsonii* a lo largo de toda su distribución reproductiva en el sureste de Estados Unidos para identificar rutas migratorias y áreas no reproductivas. Además de geo-localizadores tradicionales basados en niveles de luz, utilizamos datos de presión barométrica recolectados por geo-localizadores multi-sensor (dispositivos de presión), que combinamos con un re-análisis meteorológico global y un modelo de movimiento (GeoPressureR) para estimar ubicaciones precisas y continuas, y recopilar datos de comportamiento de vuelo previamente no disponibles. Colocamos 43 dispositivos de presión y 43 geo-localizadores de nivel de luz en machos reproductores de *L. swainsonii* en seis estados durante 2021. Recapturamos 31 individuos marcados (36%) en 2022, con 26 dispositivos utilizados para el análisis. Los geo-localizadores basados en niveles de luz ($n$ = 12) experimentaron sombreado crónico, lo que resultó en estimaciones generales de las áreas no reproductivas y escasa información durante la migración. En cambio, los dispositivos de presión ($n$ = 14) proporcionaron estimaciones precisas de las áreas no reproductivas (~10–40 km) y de todas las regiones de parada. Encontramos una división longitudinal en las distribuciones no reproductivas: los individuos reproductores occidentales del valle del Misisipi y los Ozarks de Misuri migraron a través del Golfo de México para pasar la temporada no reproductiva en la península de Yucatán; los reproductores orientales de los Apalaches y las planicies costeras del Atlántico migraron a Cuba y las Bahamas, con paradas intermedias en Florida y Georgia. Los dispositivos de presión también revelaron detalles precisos de los movimientos migratorios, incluyendo la fenología, las duraciones de vuelo y las altitudes. De manera destacada, los individuos se desplazaron significativamente más rápido y pasaron menos tiempo en las áreas de parada durante la migración otoñal que en la primaveral, en contraposición al conocimiento actual. Las rutas migratorias y las áreas de invernada de *L. swainsonii* sugieren que las poblaciones reproductoras orientales y occidentales experimentan diferentes condiciones ambientales, barreras ecológicas y antrópicas, y calidades de hábitat durante la migración.


Palabras clave: conectividad, ecología de la migración, fenología, geo-localizadores barométricos, Limnothlypis swainsonii

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## INTRODUCTION

As Nearctic–Neotropical migratory bird populations face declines from anthropogenic threats and changing climatic conditions, it is important to understand movements and habitat use through the full annual cycle (Marra et al. 2015, Rosenberg et al. 2019). Although studies of Nearctic–Neotropical migrants have historically focused on the breeding season, the nonbreeding period can have profound consequences on individual fitness, migration success, and population dynamics via habitat quality, resource availability, and carry-over effects (Marra et al. 1998, Norris 2005, González-Prieto and Hobson 2013). Furthermore, migration is an inherently risky endeavor associated with increased mortality depending on available stopover habitat and environmental conditions (Newton 2006, Cohen et al. 2014, Klaassen et al. 2014, Lok et al. 2015). Thus, understanding species-specific migratory ecology and behavior (e.g., pathways, stopover use, and timing) is crucial to enacting effective conservation strategies.

Conservation of migratory species also requires understanding migratory connectivity, the pattern and degree of spatio-temporal cohesion of populations within a species over the full annual cycle (Webster et al. 2002, Cohen et al. 2018). Species with breeding populations that remain geographically isolated during the nonbreeding season exhibit strong connectivity, whereas species with breeding populations that intermix across broad nonbreeding regions exhibit weak connectivity (Cohen et al. 2018). Connectivity can be present not only between the breeding and nonbreeding grounds, but during migration as well; if two populations travel along separate defined routes, or travel similar flyways but during different time periods, this would be considered strong connectivity (Bauer et al. 2016, Briedis and Bauer 2018, Cohen et al. 2019). As a result, highly connected populations may each experience distinct habitats, environmental conditions, and stressors during the annual cycle. This interpopulation variation can explain local population declines (e.g., Hewson et al. 2016, Kramer et al. 2018), but can only be assessed by knowing the location of individuals through the annual cycle.

One migratory species that would greatly benefit from full annual cycle tracking is *Limnothlypis swainsonii* (Swainson's Warbler), a Nearctic–Neotropical migrant songbird that breeds in dense understory vegetation of interior forests throughout the southeastern United States (Anich et al. 2020). Their limited distribution, relatively small population sizes, and secretive nature have resulted in gaps in knowledge of their ecology, making them a conservation priority species for multiple state wildlife agencies and a Partners in Flight species of concern (Rosenberg et al. 2016). *Limnothlypis swainsonii* breeding range can be broadly divided into western and eastern populations. Western populations occupy bottomland or riparian hardwood forests and young pine plantations of the Western Gulf Coast, Mississippi Alluvial Valley, and Ozark Highlands from eastern Texas to Mississippi and up to southern Missouri (Thomas et al. 1996, Graves 2002, Bassett-Touchell and Stouffer 2006, Benson et al. 2009). Eastern populations occupy bottomland forests and pine plantations of the Atlantic Coastal Plains and Piedmont from north Florida to Virginia (Bishop et al. 2012), and rhododendron (*Rhododendron* spp.) and mountain laurel (*Kalmia latifolia*) coves of the southern Appalachians from Alabama to West Virginia (Lanham and Miller 2006). Both the Appalachian and Ozark populations have historically been understudied. During the nonbreeding season, *L. swainsonii* are distributed across forested habitats of the Yucatan Peninsula, including Mexico, Guatemala, Belize, and Honduras, and the Caribbean including the Bahamas and Greater Antilles (Anich et al. 2020). Similar to their breeding habitat, wintering *L. swainsonii* appear to select contiguous forests with thick leaf litter (Graves 1998, Brunner et al.

2022a), including dry limestone forests, wet broadleaf forests, and shade coffee plantations in the Caribbean (Strong and Sherry 2001, Townsend et al. 2014) and primary and secondary native forests in the Yucatan (Oliveira et al. 2021). However, many of these habitats across these nonbreeding regions are threatened by increased conversion to unsuitable ecosystems, such as palm oil plantations and grazing lands (Galford et al. 2018, Ellis et al. 2017, Oliveira et al. 2021).

Less is known about *L. swainsonii* migratory routes and phenology, as they are notoriously difficult to detect during the nonbreeding season, including during migration (Graves 1996). They are rarely captured at banding operations during migration, and no bird banded on the breeding grounds has been recaptured in the winter (USGS Bird Banding Laboratory 2021). Limited evidence from genetic data from their nonbreeding grounds (Winker and Graves 2008) and light-level geolocator data from Louisiana and Jamaica (Brunner et al. 2022b) suggest a difference in breeding origins between eastern (i.e., Caribbean) and western (i.e., mainland Central American) nonbreeding populations. However, no range-wide study has been conducted to understand the pattern of migratory connectivity.

The challenge of tracking small (<20 g) birds away from their breeding grounds means that important information about nonbreeding distributions, habitat selection, migratory pathways, stopover sites, and migratory phenology is lacking for many species (Faaborg et al. 2010, Scarpignato et al. 2023). Archival light-level geolocators are commonly employed to track small migratory birds throughout their annual cycle by estimating latitude and longitude from sunrise and sunset times (Lisovski et al. 2020). However, location estimates can be coarse (50 to >300 km) and are susceptible to error (light sensor shading, calibration, equinox periods), especially for species such as *L. swainsonii* that occupy dark forested habitats (Lisovski et al. 2012, 2018). Recently, multisensor barometric geolocators (hereafter "pressure tags") that additionally sample atmospheric pressure alongside light levels have been developed for small birds (<20 g). Because pressure values correspond to altitude, pressure tags can identify and characterize novel migratory flight patterns (Briedis et al. 2020), effectively illuminating behavior that has been difficult to study in small nocturnal migrants. However, the most important advancement from pressure tags for studying migratory routes is linking recorded pressure data from tags to spatiotemporal barometric pressure variation across a global weather dataset and elevations to generate continuous and much more accurate location estimates throughout the individual's annual cycle (Nussbaumer et al. 2023). Pressure tags overcome many of the limitations of light-level geolocation and can now uncover complete migratory pathways, exact migratory phenology, and novel flight behaviors, thus greatly enhancing our ability to understand, and subsequently conserve, small migratory birds (Rhyne et al. 2024).

In this study, we deployed archival pressure and light-level geolocators on *L. swainsonii* from 7 field sites across their breeding range to examine their migratory routes, timing, and behavior. Specifically, our objectives were to (1) determine nonbreeding regions and migration routes; (2) qualitatively describe the pattern of migratory connectivity across breeding populations; (3) characterize and compare migratory phenology and behavior among populations and seasons; (4) compare the accuracy and information gained from barometric vs. light-level geolocators; and (5) evaluate the effects of tagging with geolocators on *L. swainsonii* return rates.

## METHODS

To cover the extent of the *L. swainsonii* breeding range, we selected field sites from 4 ecoregions in the southeastern United States. For western populations, we tagged individuals in bottomland hardwoods and pine forests of Louisiana (Frenchtown Conservation Area, Feliciana Preserve, Pearl River Wildlife Management Area, Sherburne Wildlife Management Area, Creek Park, Lee Memorial Forest, and Kendallwood Road Park), Arkansas (St. Francis National Forest), and the Missouri Ozarks (Mark Twain National Forest). For eastern populations, we tagged individuals in the Coastal Plains of South Carolina (Audubon Center and Sanctuary at Francis Beidler Forest, Silver Bluff Audubon Sanctuary, and Edisto Nature Trail) and Virginia (Great Dismal Swamp National Wildlife Refuge), and the Appalachians of Tennessee (South Cherokee National Forest), and Virginia (Breaks Interstate Park and Birch Knob Preserve).

In 2021, we located and mist-netted singing male *L. swainsonii* on breeding territories from May to June (after the migration period) using conspecific audio playback. We banded (USGS aluminum band and color band for resighting purposes, permit #22648), measured, sexed, and aged each individual and fitted them with geolocators. We deployed 43 archival barometric pressure + light-level geolocators (Intigeo BARP50Z11-7, Migrate Technology, Cambridge, UK, 0.58 g, Supplementary Material Figure 1) and 43 archival light-level geolocators (Intigeo P50Z11-7-DIP-NOT, Migrate Technology, Cambridge, UK, 0.65 g). All geolocators were programmed to record light (lux) every minute and store the average every 5 min, while pressure tags also sampled atmospheric pressure (Pa) every 30 min. We fitted geolocators using a 0.08-g field-adjustable leg-loop harness made of 0.8-mm Stretch Magic jewelry cord and aluminum crimping beads (Jirinec et al. 2021). We also banded 40 *L. swainsonii* following the same protocol but did not fit them with geolocators. This group served as a control, allowing us to test for tagging effects on apparent return rates. We returned to capture sites in 2022 to recapture both tagged and control individuals and recover geolocators.

### Pressure Analysis

To estimate nonbreeding locations from pressure data, we implemented the *GeoPressureR* package (Nussbaumer and Gravey 2022) in program R version 4.1.1 (www.r-project.org) and its pipeline (Nussbaumer et al. 2023a). Although this analysis can incorporate both light and pressure data, we used only pressure data, as light data added no additional resolution (see below). We began by classifying raw pressure (kPa) datapoints as either stationary period, migratory activity, or outlier. We identified migratory activity as irregular sharp drops in pressure values that correspond to high-altitude migratory flights; whereas stationary periods were long spans of consistent higher-pressure patterns, including breeding, nonbreeding, and stopover regions (Nussbaumer et al. 2023a; Figure 1). We examined the data to identify outliers during stationary periods, as described in the GeoPressureR manual (https://raphaelnussbaumer.com/GeoPressureManual/), including removing single data points that strayed disproportionately from the surrounding points. We also identified shifts in elevation indicating either daily or seasonal movements separate from migration. Once we had clean pressure data for each stationary

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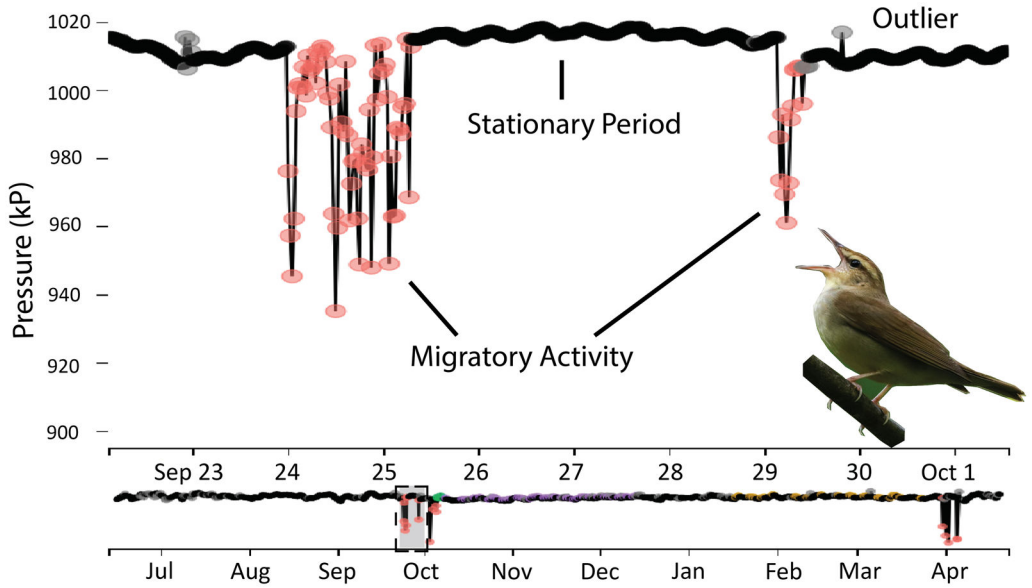



**FIGURE 1.** Pressure data labeling process from the TRAINSET tool as part of the *GeoPressureR* analysis. Migratory activity (red points) are identified by irregular sharp drops in pressure that correspond to increased altitude in flight, while stationary periods (black points) show even fluctuations. Outliers (gray points) are discarded. *L. swainsonii* photo by Martina Nordstrand.

period, we calculated minimum and maximum ground-level elevations (~5–10 m accuracy) during that period to serve as an elevational mask that excluded global positions falling outside of the range. We then compared the spatial-temporal variation in the pressure data with a global weather reanalysis dataset, ERA5 (Muñoz-Sabater et al. 2021), by calculating a mean square error (MSE) between the two datasets. Because these weather reanalysis data are available every hour, we were able to estimate locations for short stationary periods (i.e., migratory stopover sites) even <1 day. Using the elevational masks and the pressure MSE, we generated log-linear pooled likelihood maps on 10 × 10 km gridded cells. We further refined our stationary estimates using a hidden Markov movement model (Nussbaumer et al. 2023b). By incorporating defined flight metrics (departure and arrival time, and flight duration) and high-resolution ERA5 windspeed data recorded at multiple atmospheric altitudes, we calculated transition probabilities between each possible likelihood map cell. Essentially, this further informed our estimates by providing a realistic flight radius between two stationary periods and accounted for the effects of wind drift and wind assistance during those flights, which greatly influence the direction and speed of migratory birds (Werfeli et al. 2022). The movement model is more important for short-term migratory stopover estimates, whereas long-term estimates for the nonbreeding grounds gain little improvement due to the original higher confidence. From the combination of our likelihood maps and movement model, we produced a marginal probability map of each stationary period at a 10 × 10 km resolution and the most likely migration trajectory connecting each stationary period.

Pressure data also allowed us to delineate fine-scale migratory phenology (migratory departure, duration, and arrival) and characterize migratory flight metrics and behaviors across seasons and deployment sites. This included migratory speed (km day$^{-1}$), total distance, maximum and mean flight altitudes, proportion of time spent at stopover sites, and relationships between departure time and flight distance and duration. Then we modeled total distance by mean and maximum altitude gained in flight using a simple linear model. To evaluate the relationship of sunset to migratory flight initiation, we modeled flight distances and durations across departure times relative to sunset recorded by geolocators using a generalized additive model (GAM) with the R package *mgcv* (Wood 2011) with each individual bird as a random effect.

## Light Analysis

For geolocators that only recorded light-levels, we analyzed data with R packages *TwGeos* (Lisovski et al. 2016) and *SGAT* (Wotherspoon et al. 2013) following the outlined procedure by Lisovski et al. (2020). Using raw light values (lux), we annotated twilight events (sunrise and sunset) each day based upon a threshold value (0.5), then conservatively edited them with the *twilightEdit* function. We used an "in-habitat" calibration period between June 15 and August 15 (40–60 days), in which tagged individuals were at their known deployment location, allowing an error distribution of twilight times to be generated. Although it can be difficult to distinguish migratory periods with light data, we identified conspicuously high light values that correspond to diurnal migratory events in several individuals, which should only occur when crossing large geographic barriers that inhibit stopover (i.e., the Gulf of Mexico, Brunner et al. 2022b). We estimated a single nonbreeding location from light data between October 15 and February 15, when light analysis reaches maximal accuracy due to temporal separation from the equinoxes. We estimated nonbreeding locations with the *SGAT* package that employs both the "threshold" and "curve" methods to create an initial path, which was then fed into a Monte Carlo Markov Chain (MCMC) model complete with land masks, priors, and defined known locations. Final estimates result in a raster heat map of predicted probabilities. Due to the coarse nature of light-level estimation (50 to >200 km), we chose to preserve the longitudinal integrity of the light data by reducing land mask strength that may otherwise pull centroids away from the more accurate longitudinal

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estimate, resulting in some data-deficient location estimates to occur in unlikely regions (e.g., over water).

In addition, we compared the accuracy of barometric and light-level geolocation based on location estimates from all pressure tags that recorded both measurements. We based this comparison on location estimates generated from July 1 to August 1, from individuals on or close to their breeding territories. No prior location information is incorporated into raw pressure estimates, which become more accurate as pressure samples accumulate. Initial location data are necessary for light estimate calibration. After estimating locations using both light and pressure data separately, we then determined the distance between the estimated centroids and the known deployment site. We conducted a t-test between these data to evaluate any difference.

Migratory connectivity (MC) of a population can be quantified (Cohen et al. 2018). However, we do not feel our data are fit for calculating a MC value because (1) our dataset has a low sample size, and some points are poor-quality light-level estimates ($n = 26$, see Results); (2) is missing a key field site (Great Dismal Swamp) from the Coastal Plains population; (3) is only representative of one year rather than multiple years; (4) would rely on relative abundance estimates from a poorly detected species; and (5) incorporates unequal error distribution from pressure analyses that are not integrated in the *MigConnectivity* package (but see Hostetler et al. 2025). The combination of these factors produces spatial and temporal biases that likely result in a misleading connectivity estimate (Vickers et al. 2021). Thus, we did not calculate MC but instead describe connectivity qualitatively based on the pattern of breeding and nonbreeding locations.

## RESULTS

We recovered 31 geolocators (36% recapture, Table 1). Two geolocators from Virginia and Tennessee malfunctioned during deployment and three geolocators from the Great Dismal Swamp (Virginia) were stolen while in-transit prior to data download, resulting in 26 usable tags for analysis (14 pressure and light-level, 12 light-level only) with complete data from July 2021 through April–May 2022. We resighted or recaptured 15 of the 40 control banded individuals (38% return), thus finding no evidence of an effect of geolocator tagging on apparent return rates ($P = 0.839$, $\chi^2 = 0.041$).

Data from pressure tags allowed us to estimate accurate and refined nonbreeding sites, including stopover locations used for <1 day, illustrating a continuous migratory pathway for 14 individuals (Figure 2). Light-level data, which experienced chronic shading, provided broad (~100–200 km) estimates for nonbreeding grounds of 12 individuals (Figure 3, Supplementary Material Figure 2). Overall, we found *L. swainsonii* exhibited longitudinal migratory connectivity, meaning a separation between western and eastern breeding populations and their respective nonbreeding regions (Figure 3). Western populations spent the nonbreeding season in the Yucatan peninsula ($n = 10$), including coastal and inland sites of northern Guatemala and the Mexican states of Campeche, Chiapas, and Quintana Roo. All western individuals flew directly across the Gulf of Mexico for both fall and spring migration, traveling an average of 1,572 ± 308 km for fall migration and 1,808 ± 307 km for spring. Tag CB627 was an exception to this pattern during spring migration, when it was displaced eastward by strong winds during spring migration, resulting in a loop-migration from the Yucatan to the western coast of Florida and back across the upper Gulf coast to its breeding site in Louisiana (Supplementary Material Figure 3). Eastern populations spent the nonbreeding season in distinctly different locations: Coastal Plains breeders in Cuba ($n = 3$) and the Bahamas ($n = 1$), and Appalachian breeders in central and western Cuba ($n = 12$). All eastern individuals used Florida as a pathway during fall and spring migrations to reach Cuba, stopping in both inland and coastal sites. Appalachian breeders traveled 1,633 ± 222 km in fall and 1,575 ± 117 km in spring, whereas Coastal Plains breeders traveled 1,328 ± 309 km and 1,164 ± 76 km in the fall and spring, respectively. We noted a commonly shared stopover site during spring migration for 6 of the 8 eastern breeders in the Okefenokee Swamp region along the Georgia and Florida border. Full data for all 14 pressure tags are available through the GeoLocator Data Package (Rhyne 2025).

When comparing light and pressure data from the same tags at the capture location, we found location estimates derived from pressure (median = 28.4 km away from known capture location) to be significantly more accurate than light (median = 165.7 km, $P = 0.001$, $t = -4.297$, Supplementary Material Figure 4), even though Appalachian breeders exhibited daily elevational movements that likely reduced the accuracy of pressure data. Although we could not compare exact locations between the two methods on the nonbreeding grounds, the differences were further exaggerated because individuals experienced heavier shading than on the breeding grounds, making light-level data even less accurate in the nonbreeding season.

Pressure tags ($n = 14$) also revealed exact departure and arrival dates of individuals, as well as time spent in-flight and at stopover sites (Table 2, Supplementary Material Figure 5), whereas light data only revealed diurnal migration events (Supplementary Material Table 6). From pressure data, we know that all individuals departed from their breeding grounds between September 4 and October 16, with fall migration

**TABLE 1.** Geolocator deployment locations and recovery rates of *L. swainsonii*.

| Population | State | Geolocators deployed | Geolocators recovered (*% recovered*) | Control individuals | Control return (*% control return*) |
|---|---|---|---|---|---|
| Western | Arkansas | 1 | 0 (*0%*) | – | – |
| | Louisiana | 17 | 7 (*41%*) | 15 | 8 (*53%*) |
| | Missouri | 10 | 3 (*30%*) | 2 | 0 (*0%*) |
| Coastal Plains | South Carolina | 10 | 4 (*40%*) | – | – |
| | Virginia | 14 | 3 (*21%*) | 12 | 3 (*25%*) |
| Appalachian | Tennessee | 18 | 5 (*28%*) | 8 | 2 (*25%*) |
| | Virginia | 16 | 9 (*56%*) | 3 | 2 (*66%*) |
| Total | | 86 | 31 (*36%*) | 40 | 15 (*38%*) |

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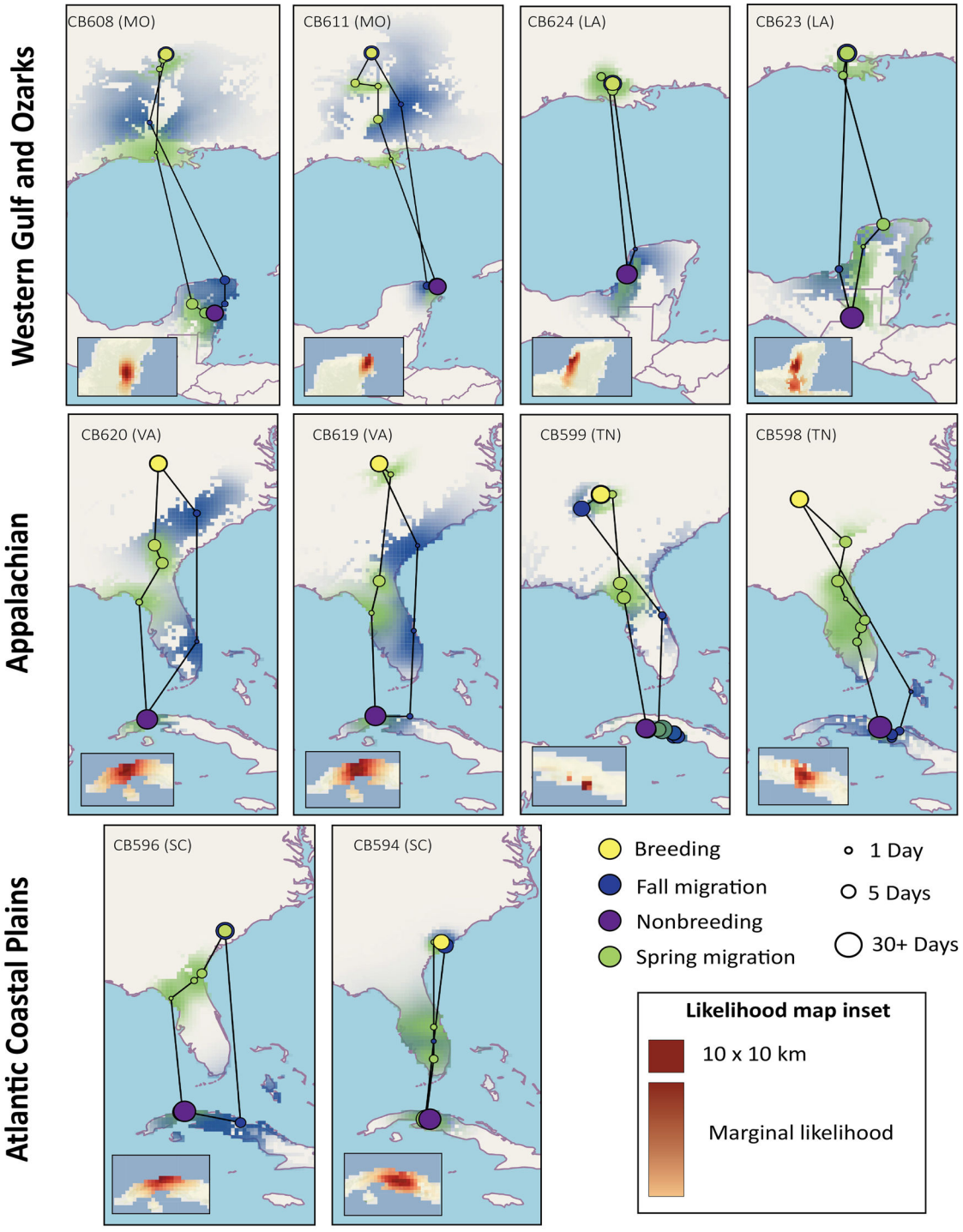



**FIGURE 2.** Complete migratory trajectories derived from pressure tags, arranged by breeding populations (left label). Maps include centroid and 80% probability distribution for each stationary period, including nonbreeding site (purple), and fall (blue) and spring (green) migration stopover sites. The size of a point indicates the relative duration of stay. The inset for each tag trajectory is the likelihood distribution map of the long-term nonbreeding estimate in a 10 × 10 km grid. Actual location is likely within <50 km of highest likelihood cell(s), and error is distributed irregularly across geographic features.

lasting between 1.3 and 11.3 days (mean = 3.9 ± 2.8 SE). Spring migration commenced between March 11 and April 18, with migration lasting between 3.4 and 27.9 days (mean = 11.6 ± 6.9 SE). Overall, *L. swainsonii* traveled faster during fall migration (550 km day$^{-1}$) than in spring (184 km day$^{-1}$) ($P < 0.001$, $t = 4.6$, df = 15.8), spent proportionately less time at stopover sites in the fall (51%, mean = 2.4 days ± 0.7 SE) than during the spring (81%, mean = 10.2 days ± 1.8 SE, $P < 0.001$, $\chi^2 = 19.8$), but did not travel significantly different distances between seasons ($P = 0.63$, $t = -0.49$, df = 25.7).

Pressure tags recorded 115 individual migratory flights of a combined 947 hr of in-air data. Flights ranged from 1.25 to 30.75 hr (mean = 8.3 ± 6.4 SE), traveling a mean of 389.5 ± 376.8 km per flight. The maximum distance recorded in a single flight was ~1,410 km by CB621 (Louisiana) over the Gulf of Mexico. Flight duration and distance traveled were

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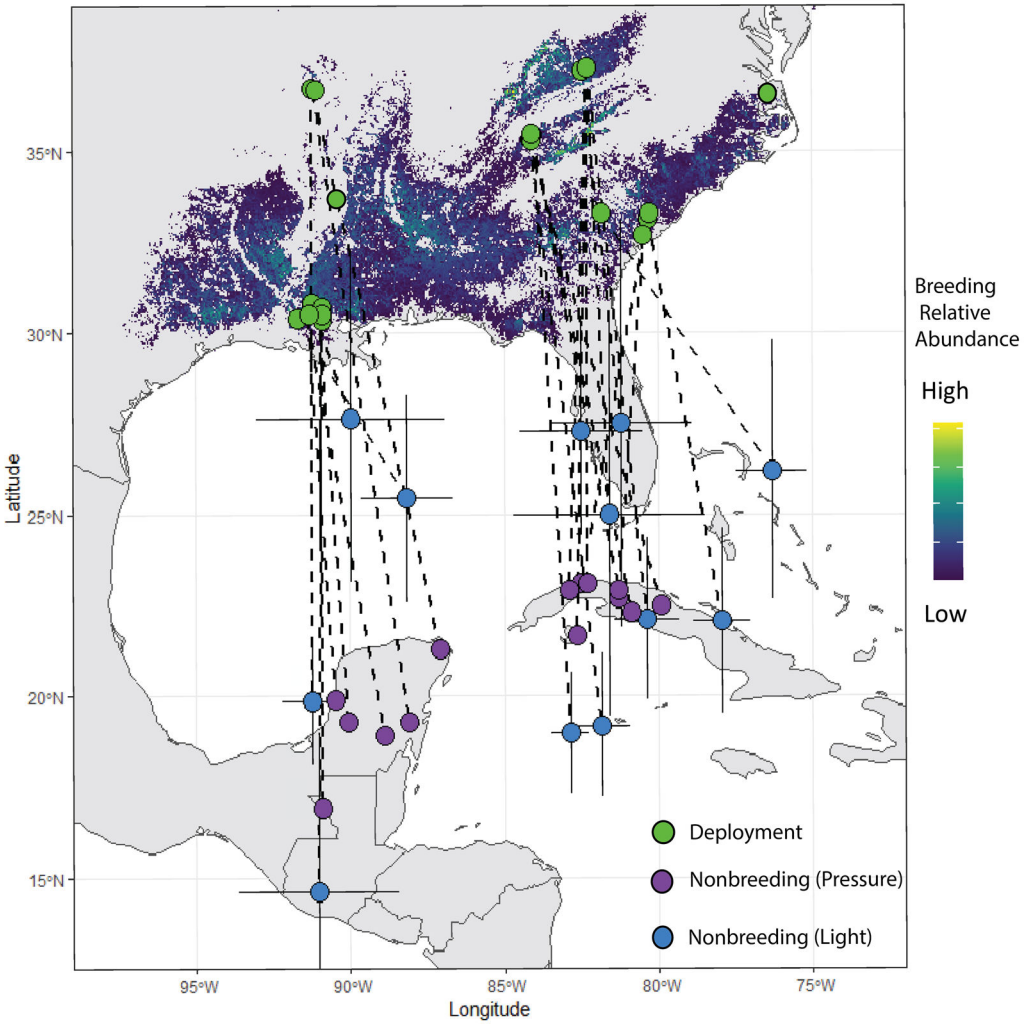



**FIGURE 3.** Nonbreeding locations of tracked *L. swainsonii*. Tag deployment sites (green) are connected to the individual's estimated nonbreeding site either by more accurate pressure tags (purple) or less accurate light-level tags (blue, see Methods) with SE bars, overlaid atop *L. swainsonii* eBird breeding abundance map (Fink et al. 2023). These connections do not incorporate the migratory route taken. These range-wide tracks demonstrate a separation of eastern and western populations.

positively correlated ($r = 0.914$, $P < 0.01$), allowing us to use distance as a predictor for other flight analyses. Flight altitudes varied depending on the distance traveled in flight. The mean altitude per flight was 413 ± 270 m, with a maximum altitude per flight of 881 ± 557 m. We found a significantly positive linear relationship between distance and both mean flight altitude ($r = 0.225$, $P = 0.015$) and maximum flight altitude ($r = 0.879$, $P < 0.01$). Individuals migrated at significantly higher altitudes in the spring than in the fall ($P = 0.005$, $t = -3.1$, df = 26). The highest flight altitude recorded was 2,692 m by CB620 (VA) flying over the Appalachians during spring migration. Flight departures averaged 91 min after sunset (range: 0–362 min). Using a GAM, we found a non-linear relationship between departure time after sunset and flight distance (46% deviance explained): individuals leaving for longer flights departed earlier, but departure time was variable when completing shorter flights (Figure 4).

## DISCUSSION

By tracking *L. swainsonii* throughout their annual cycle, we documented complete migratory trajectories from 14 barometric geolocators, and general nonbreeding regions or longitudes from an additional 12 light-level geolocators. As in previous studies, we demonstrated no apparent effect of tags on return rates (Anich et al. 2009, Brlík et al. 2020). Eastern breeders from the Appalachians and Atlantic Coastal Plains migrated to the Caribbean while western breeding birds from the Mississippi Alluvial Valley and Ozark Highlands migrated to the Yucatan Peninsula. Individuals migrated faster in the fall than in the spring due to longer spring stopovers. Detailed behavioral data from pressure tags showed that migrating individuals often departed >1 hr after sunset, especially for relatively short (<100 km) flights. Eastern breeders from the Appalachians and Atlantic Coastal Plain spent the nonbreeding season across Cuba, both along the lower-elevation coast and within the interior mountains, and in the Bahamas. All eastern breeders migrated through and stopped in peninsular Florida, which is a well-documented migratory flyway for migrant passerines, especially parulids (Lowery and Newman 1966, Lott et al. 2006).

The Okefenokee Swamp, one of the largest remaining intact bottomland hardwood wetlands in the U.S. in southern Georgia and northern Florida, served as a frequent stopover site during

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**TABLE 2.** Results derived from 14 recovered pressure tags on *L. swainsonii*.

| | | | | Fall migration | | | | | Spring migration | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Tag | Breeding state | Nonbreeding region | Lat–long estimate | Depart–arrive | Total distance (km) | Flight hours | Stopover hours | Mean altitude (m) | Depart–arrive | Total distance (km) | Flight hours | Stopover hours | Mean altitude (m) |
| CB621 | LA | Northern Guatemala | (16.9, –90.9) | Oct 4–Oct 7 | 1,531 | 37.50 | 19.25 | 419 | Apr 4–Apr 10 | 1,703 | 40.25 | 85.00 | 649 |
| CB627 | LA | Quintana Roo, Mexico | (18.9, –88.9) | Sep 22–Sep 24 | 1,312 | 26.50 | 5.75 | 453 | Mar 22–Apr 6 | 2,232 | 58.15 | 283.10 | 361 |
| CB624 | LA | Campeche, Mexico | (19.7, –90.5) | Sep 22–Sep 24 | 1,209 | 26.75 | 25.00 | 350 | Mar 28–Apr 2 | 1,407 | 25.25 | 75.75 | 741 |
| CB623 | LA | Northern Guatemala | (16.7, –90.5) | Sep 22–Sep 24 | 1,535 | 28.25 | 4.50 | 428 | Mar 22–Mar 29 | 1,570 | 39.00 | 132.75 | 665 |
| CB608 | MO | Quintana Roo, Mexico | (19.3, –88.1) | Oct 3–Oct 11 | 2,036 | 48.00 | 127.25 | 394 | Mar 11–Apr 8 | 2,053 | 34.90 | 635.85 | 523 |
| CB611 | MO | Quintana Roo, Mexico | (21.3, –87.1) | Sep 22–Sep 27 | 1,812 | 45.25 | 56.50 | 402 | Apr 10–Apr 20 | 1,883 | 33.75 | 189.25 | 534 |
| CB596 | SC | Western Cuba | (23.1, –82.5) | Sep 23–Oct 5 | 1,546 | 50.00 | 221.25 | 232 | Mar 30–Apr 5 | 1,218 | 20.50 | 104.75 | 527 |
| CB594 | SC | Western Cuba | (22.7, –81.3) | Sep 23–Sep 25 | 1,109 | 25.75 | 25.50 | 223 | Apr 1–Apr 5 | 1,110 | 25.75 | 55.00 | 380 |
| CB598 | TN | Central Cuba | (22.5, –79.9) | Sep 22–Sep 29 | 1,722 | 34.50 | 116.00 | 433 | Mar 24–Apr 11 | 1,439 | 43.25 | 379.25 | 439 |
| CB601 | TN | Western Cuba | (22.9, –81.3) | Sep 21–Sep 25 | 1,195 | 36.00 | 64.00 | 603 | Mar 22–Apr 11 | 1,453 | 31.00 | 434.50 | 434 |
| CB599 | TN | Central Cuba | (22.1, –80.3) | Sep 22–Sep 24 | 1,606 | 33.75 | 18.25 | 417 | Mar 30–Apr 13 | 1,580 | 30.00 | 287.50 | 621 |
| CB616 | VA | Western Cuba | (21.7, –82.6) | Oct 16–Oct 19 | 1,765 | 29.00 | 25.00 | 326 | Apr 5–Apr 17 | 1,755 | 31.00 | 239.00 | 418 |
| CB619 | VA | Western Cuba | (23.1, –82.3) | Sep 23–Sep 27 | 1,768 | 39.00 | 41.00 | 287 | Apr 5–Apr 15 | 1,598 | 28.50 | 192.00 | 679 |
| CB620 | VA | Western Cuba | (22.9, –82.9) | Sep 30–Oct 4 | 1,740 | 35.00 | 64.75 | 661 | Apr 5–Apr 21 | 1,626 | 28.50 | 342.75 | 773 |

Each row provides migratory metrics of a tagged individual that includes the deployment state, long-term nonbreeding location, migration departure and arrivals, distance flown, flight durations, stopover durations, and flight altitude.

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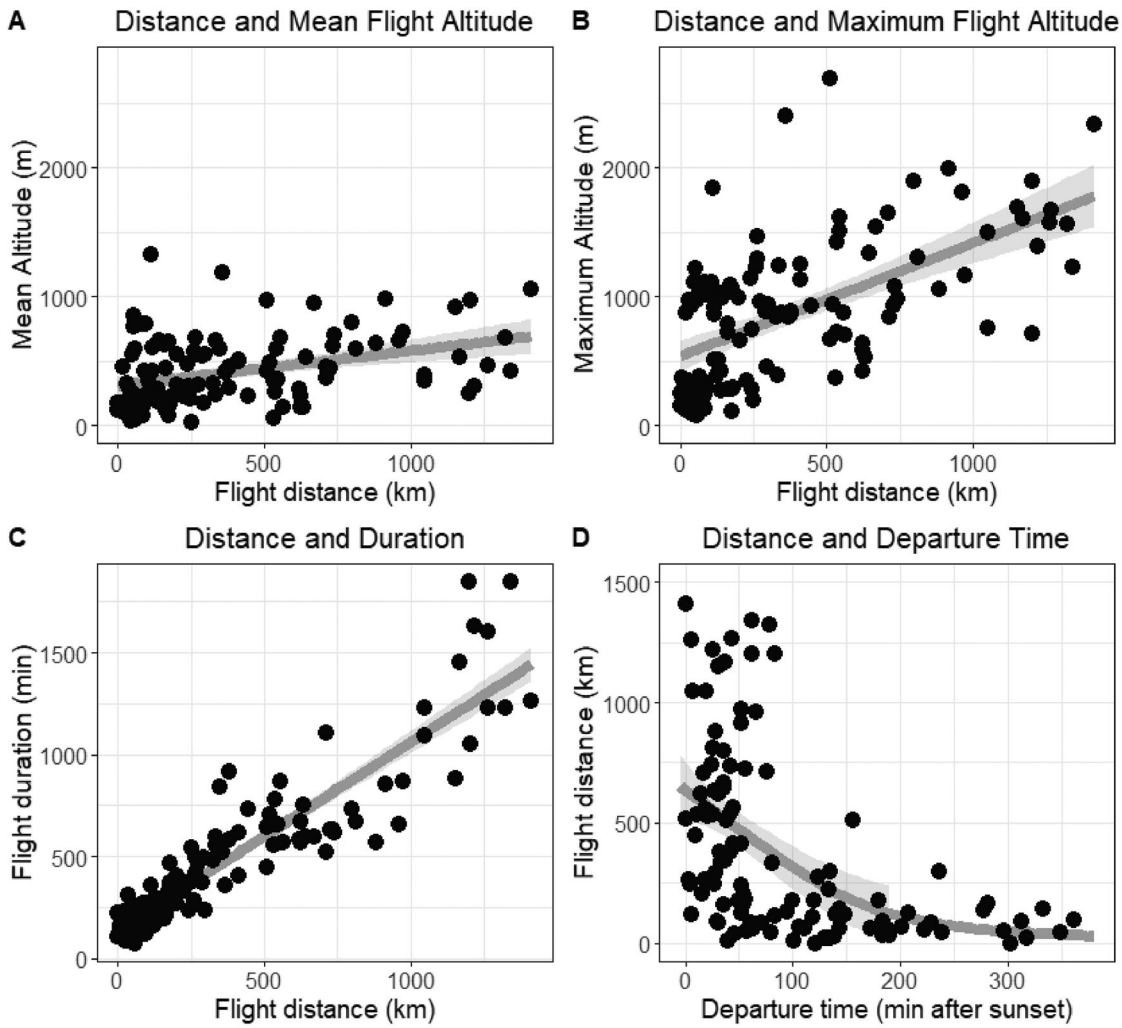



**FIGURE 4.** Individual flight metrics and relationships. Data from 115 recorded migratory flights were used to model: linear regressions of (A) mean altitude and flight distance; (B) maximum altitude and flight distance; (C) flight duration and flight distance; and (D) a generalized additive model (GAM) of flight departure time after sunset and flight distance.

spring migration for eastern *L. swainsonii*. Almost all returning eastern individuals stopped in Okefenokee for extended periods (3–10 days) before reaching their breeding grounds, likely indicating that it is high quality stopover habitat that provides important resources for refueling. However, the Okefenokee Swamp currently faces pressures from mineral extraction operations, which may alter forest habitat regimes and resource availability, presenting further threats for these migrants (Jackson 2022).

Conversely, western-breeding *L. swainsonii* from the Mississippi Alluvial Valley and Ozarks, flew across the Gulf of Mexico to spend the nonbreeding season in the Yucatan Peninsula coastline and interior. Based on pressure data, *L. swainsonii* complete continuous trans-Gulf flights of 14 to 30 hr, sometimes departing or arriving >50 km inland. Although a trans-Gulf route is the shortest, most efficient flight path to reach the Yucatan Peninsula, it requires an extensive over-water flight, which increases risk of exposure to extreme weather, dehydration, and exhaustion (Deppe et al. 2015). By following a direct trans-Gulf migration route, rather than a circum-Gulf route along eastern Texas and Mexico, western individuals used fewer stopover sites than their eastern counterparts. Many individuals used the northern Yucatan and Campeche coastline for both fall arrival stopover and spring staging prior to crossing the Gulf. Even so, two individuals from Missouri did not use Louisiana coastline for either spring or fall migration, instead departing and arriving inland, often >200 km away from the coastline. Only two individuals from Louisiana used coastal habitat for spring stopover.

Patterns in nonbreeding grounds and migratory routes from across their breeding range suggest *L. swainsonii* exhibit some degree of connectivity along a longitudinal divide. Brunner et al. (2022b) found similar patterns of connectivity for *L. swainsonii* from wintering and breeding tagging sites. Thus, eastern and western breeding populations experience different environmental conditions, habitat availability, and migratory obstacles throughout their annual cycle. Continued tracking efforts of *L. swainsonii* that can provide a higher sample size and spatial-temporal variation will provide more suitable data from which an ecologically useful MC value can be calculated.

Given the pattern of parallel connectivity, we speculate that easternmost *L. swainsonii* from the Great Dismal Swamp, for which our data were lost, likely spent the nonbreeding season in Jamaica or eastern Cuba, as suggested by Brunner et al. (2022b). Because we did not sample breeding populations from Alabama, Mississippi, or Georgia it is unclear where eastern

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and western nonbreeding populations diverge in their migratory patterns. Far western breeders in eastern Texas may migrate circum-Gulf, as historical records indicate spring migration of *L. swainsonii* on the southern Texas coastline (Meanley 1971). Additionally, we only tracked males, as females were rarely captured, and would be difficult to recapture given their reluctance to approach song playback. Most migration research is sex-biased towards males for similar reasons. However, it is critical to understand sex differences in migration timing (Briedis et al. 2019), migratory routes (Rokitnicki 2023), and nonbreeding distributions (Stouffer and Dwyer 2003, Bennett et al. 2019), as this impacts demography. Devices that do not require recapture, particularly the Motus automated radio telemetry network, might be better tools to investigate migration in females for *L. swainsonii* and other small birds.

Similar patterns of parallel connectivity have been documented in several other Neotropical migrants that also winter throughout northern Central America and the Caribbean. For example, *Seiurus aurocapilla* (Ovenbird; Hallworth et al. 2015, Haché et al. 2017), *Setophaga ruticilla* (American Redstarts; Norris et al. 2006), and *Dumetella carolinensis* (Gray Catbird; Ryder et al. 2011) all exhibit similar east–west divisions of nonbreeding populations. *Protonotaria citrea* (Prothonotary Warbler) showed differences in migration routes between eastern and western populations similar to *L. swainsonii*, but all spent the nonbreeding season within the same region (Tonra et al. 2019). Strong migratory connectivity is associated with differential population trends in other threatened parulids, including *Vermivora chrysoptera* (Golden-winged Warbler) and *Setophaga cerulea* (Cerulean Warbler), as some breeding populations migrated to well-protected regions while other breeding populations migrated to areas with higher rates of habitat loss (Jones et al. 2008, Kramer et al. 2018).

Our results also revealed new information about migratory phenology. All of our tracked *L. swainsonii* departed the breeding grounds from late September to early October. In contrast, departure from the nonbreeding grounds was spread more widely from mid-March to mid-April. Appalachian breeders all departed from the nonbreeding grounds later than the other populations. We did not observe a difference in migration timing between western and eastern *L. swainsonii*, unlike Brunner et al. (2022b). Males with seasonally increased testosterone levels are thought to migrate with minimal stops during spring migration to facilitate territory acquisition, compared to a slower migration in the fall (Tøttrup et al. 2012, La Sorte and Fink 2017; but see Cooper et al. 2017, Raybuck et al. 2022). Contrary to this idea, *L. swainsonii* spent significantly less time migrating in the fall than in the spring. Most individuals returned to their nonbreeding grounds in 3–10 days, compared to 1–3 weeks returning to their breeding grounds, with far more time at stopovers. Given that *L. swainsonii* spend ~6 months of their annual cycle on the nonbreeding grounds, our results may indicate the importance of quickly occupying high-quality nonbreeding territories. However, migration speeds are variable based on conditions, such as temperature, food abundance, rainfall, individual fitness, and distances (Cohen et al. 2015, La Sorte and Fink 2017), and therefore this pattern may instead be due to annual variation not assessed in our single-year study. Future studies should evaluate how phenology varies from year to year.

Along with location and phenological information, pressure data also allowed us to inspect metrics of each migratory flight between two estimated locations with unprecedented detail, including approximate departure times and flight altitudes. These data can help us better understand the mechanisms associated with birds initiating and sustaining nocturnal flight, something that is typically very difficult to study. For example, we found varying nightly departure times depending on how far an individual flew that night, with long flights (over ~10 hr) requiring take-offs within an hour after sunset. In contrast, individuals making shorter flights departed 30 min to 5 hr after sunset and would often finish 1–3 hr prior to dawn. This departure pattern is reflected in recent studies that suggest greater variation in departure times than had previously been thought (Müller et al. 2016, Cooper et al. 2023). Rather than being strictly cued by sunrise and sunset, migrating birds likely vary their departure and flight duration due to individual body condition and favorable flight conditions (i.e., tailwinds, temperature, and cloud cover, Deppe et al. 2015, Clipp et al. 2021). Atmospheric and weather conditions also likely dictate flight altitudes (Kemp 2013, Liechti et al. 2018), which we observed to fluctuate by hundreds of meters during individual flights, particularly for long trans-gulf flights, similarly to Bowlin et al. (2015). For shorter, over-land migratory flights, *L. swainsonii* regularly flew at ~100–300 m. Pressure tag data from additional species can reveal if *L. swainsonii* fly unusually low, which could account for their higher vulnerability to structure collisions (radio towers, buildings, etc.) compared to other parulids (Longcore et al. 2013).

Our implementation of pressure tags produced extraordinary details of migration for *L. swainsonii*, a species with little migratory information. By identifying important nonbreeding regions, our data can help inform efforts to conserve continuous forest tracts that are important for *L. swainsonii* across their annual cycle. Applying a similar research program to other migratory passerines could lead to a much more complete understanding of migration ecology and nonbreeding distributions to inform conservation efforts, including not just identifying important habitat, but also determining and potentially mitigating risks to migratory species, particularly with respect to infrastructure. Pressure data also have the potential to address a wide range of questions central to bird movements and migration, including flight behavior, physiology, daily activity, and seasonal movements.

## Supplementary material

Supplementary material is available at *Ornithological Applications* online.

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## Ethics statement

All birds were handled properly with required federal, state, and local permits and permissions.

## Conflict of interest statement

The authors have no competing interests.

## Author contributions

G.R. (conceived and designed study, conducted field work, analyzed data, and wrote paper), A.P. (conducted field work), L.B. (conducted field work), T.B. (conducted field work, substantially edited paper), D.B. (conducted field work, substantially edited paper), J.T. (conducted field work), M.J. (conducted field work, substantially edited paper), D.R. (conducted field work), R.N. (assisted in data analysis), P.S. (designed study, wrote paper, contributed substantial resources).

## Data availability

All raw geolocator data and analyses are publicly available at Rhyne et al. (2026).

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